Investigating the Efficacy of Option-Conditional Value Prediction in Reinforcement Learning

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Introduction

In model-based reinforcement learning (RL), building an accurate observation-prediction model for a complex and/or stochastic domain can be an incredibly difficult task. Oh et al.'s **Value Prediction Network (VPN)** is a deep RL architecture that circumvents the challenge of observation-prediction with an *option-conditional value prediction model*, generating the value/reward of future states without explicitly predicting future observations [1].

In this project, we investigate the efficacy of option-conditional value prediction in general RL by adapting VPN for tabular environments as well as by implementing the algorithm as in [1], using a combination of temporal-difference search (TD search) [2] and n-step Q-learning for training [3].

General Architecture

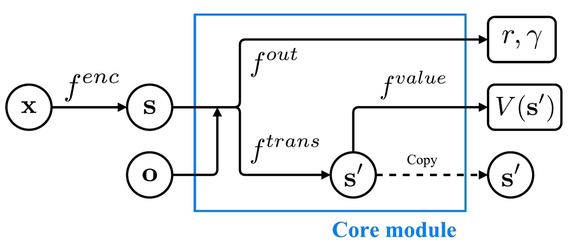
VPN is composed of four main modules, parameterized by $\theta = \{\theta^{enc}, \theta^{value}, \theta^{out}, \theta^{trans}\}$:

→ **Encoding** ($f_\theta^{enc} : \mathbf{x} \mapsto \mathbf{s}$) maps the observation $\mathbf{x}$ to the abstract state $\mathbf{s}$.

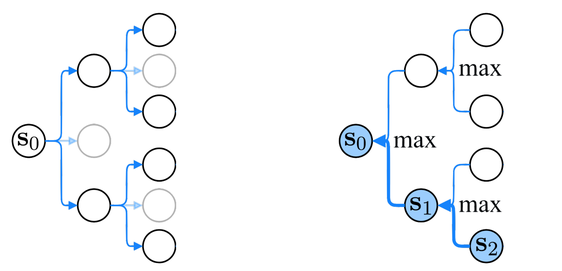
→ **Value** ($f_\theta^{val} : \mathbf{s} \mapsto V_\theta(\mathbf{s})$) estimates the value of the next abstract state, $V_\theta(\mathbf{s}')$.

→ **Outcome** ($f_\theta^{out} : \mathbf{s}, \mathbf{o} \mapsto r, \gamma$) predicts the reward r and discount factor γ corresponding to some action $\mathbf{o}$ in abstract state $\mathbf{s}$.

→ **Transition** ($f_\theta^{trans} : \mathbf{s}, \mathbf{o} \mapsto \mathbf{s}'$) transforms an abstract state-action pair $(\mathbf{s}, \mathbf{o})$ to the next abstract state $\mathbf{s}'$.



Planning



(a) Expansion (b) Backup

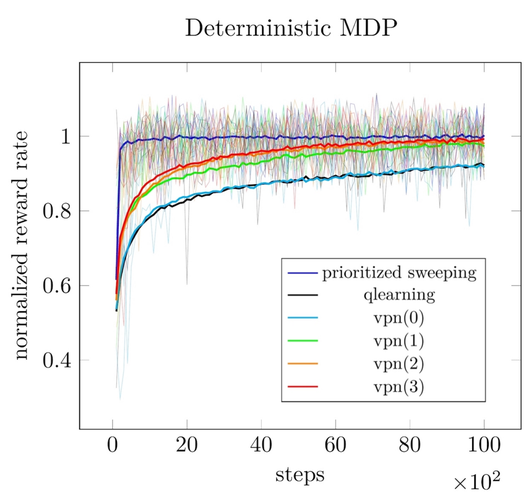
We use the following algorithm to find the Q-values of abstract state-option pairs with d-step planning:

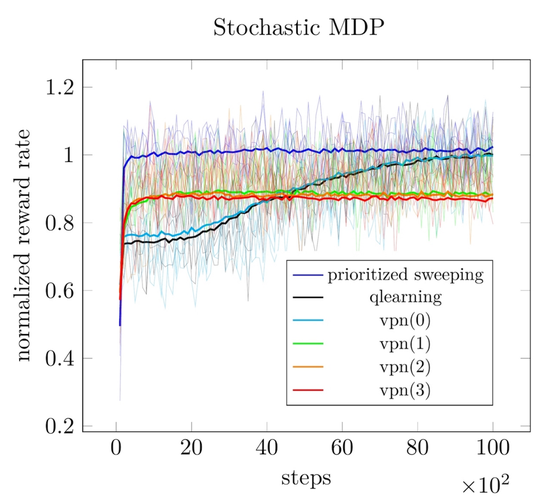
Algorithm 1 Q-value from d-step planning

function Q-PLAN($\mathbf{s}, \mathbf{o}, d$)
  $r, \gamma, V(\mathbf{s}'), \mathbf{s}' \leftarrow f_\theta^{core}(\mathbf{s}, \mathbf{o})$
  if d=1 **then**
    return $r + \gamma V(\mathbf{s}')$
  end if
  $\mathcal{A} \leftarrow$ (b-best options based on $Q^1(\mathbf{s}', \mathbf{o}')$)
  for $\mathbf{o}' \in \mathcal{A}$ **do**
    $q_{o'} \leftarrow$ Q-PLAN($\mathbf{s}', \mathbf{o}', d-1$)
  end for
  return $r + \gamma[\frac{1}{d}V(\mathbf{s}') + \frac{d-1}{d}\max_{\mathbf{o}' \in \mathcal{A}} q_{\mathbf{o}'}]$
end function

VPNTabular

In tabular environments, we can track values, rewards, and Q-values directly, sampling over past experiences to predict future states. Thus, learning is made trivial and we are able to evaluate VPN's planning scheme at face value:





VPNTabular*

We can correct for VPNTabular's performance saturation in stochastic environments by increasing sampling frequency during Q-value calculation.

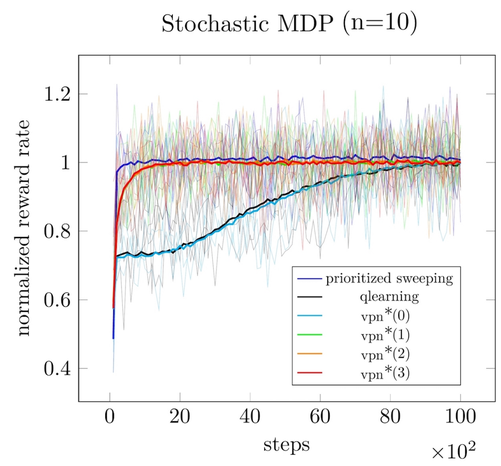
In VPNTabular*, we replace

$$o_t \leftarrow \text{argmax}_o Q^d(s_t, o_t)$$

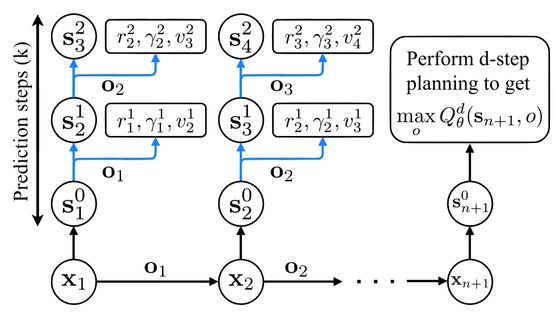
with

$$o_t \leftarrow \text{argmax}_o \left[\frac{1}{n}\Sigma_{i=1}^n Q^d(s_t, o_t)\right],$$

eliminating any saturation effects.



Learning



Given an n-step trajectory, k-step predictions are defined as follows:

$$s_t^k = \begin{cases} f_\theta^{enc}(\mathbf{x}_t), & \text{if } k = 0. \\ f_\theta^{trans}(\mathbf{s}_{t-1}^{k-1}, \mathbf{o}_{t-1}), & \text{if } k > 0. \end{cases}$$

$$v_t^k = f_\theta^{value}(\mathbf{s_t}^k)$$

$$r_t^k, \gamma_t^k = f_\theta^{out}(\mathbf{s_t}^{k-1}, \mathbf{o}_t).$$

Next, we aggregate rewards:

$$R_t = \begin{cases} r_t + \gamma R_{t+1}, & \text{if } t \leq n. \\ max_{\mathbf{o}} Q_{\theta-}^d(\mathbf{s}_{n+1}, \mathbf{o}), & \text{if } t = n+1. \end{cases}$$

The loss corresponding to the full trajectory is then given by:

$$\mathcal{L} = \Sigma_{i=1}^n \left(\Sigma_{j=1}^k (R_i - v_i^j)^2 + (r_i - r_i^j)^2\right).$$

Conclusion

→ Performance-wise, in stochastic, tabular environments, option-conditional value prediction benefits immensely from averaging in planning.

→ When averaging is implemented during planning, tabular option-conditional value prediction performs better in stochastic environments than deterministic ones, converging to the optimal policy nearly as quickly as prioritized sweeping.

References

[1] Junhyuk Oh, Satinder Singh, and Honglak Lee. Value prediction network. In *Advances in Neural Information Processing Systems*, pages 6118–6128, 2017.

[2] David Silver, Richard S Sutton, and Martin Müller. Temporal-difference search in computer go. *Machine learning*, 87(2):183–219, 2012.

[3] Richard S Sutton and Andrew G Barto. *Introduction to reinforcement learning*, volume 135. MIT press Cambridge, 1998.

Acknowledgements

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